Tran Quoc Truong

Product Engineer | Backend Systems & Applied AI

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Summary

Backend-focused Product Engineer building Python services and applied AI systems for document intelligence, retrieval, code intelligence, and agent evaluation.

My quality engineering background strengthens how I design clear domain boundaries, reliable asynchronous workflows, and verifiable AI-enabled products.

Experience

FPT Software, Automation Tester (Applied AI & Backend Engineering) — Ho Chi Minh City, Vietnam
Dec 2024 – present
1 year 8 months

- Designed backend, retrieval, and code-intelligence components for IQP, an enterprise Quality Engineering platform.
- Built AI-assisted workflows for document analysis, test-case generation, defect analysis, and change-impact assessment.
- Developed reusable automation across Web, mobile, API, and database layers for complex banking and payment workflows.
- Improved testing efficiency through reusable validation assets, automation utilities, and traceable quality workflows.

FPT Software Academy, Full-Stack Developer Intern — Ho Chi Minh City, Vietnam
Sept 2024 – Dec 2024
4 months

- Built the initial version of a Django learning and assessment platform with structured course, quiz, enrollment, and role-based workflows.
- Integrated a separate FastAPI face-detection service for AI-assisted exam monitoring.
- Containerized and deployed the multi-service application using Docker Compose, Nginx, and Cloudflare Tunnel.

Featured Projects

Omni-Agent — Applied AI Platform

A project-centric platform for versioned document knowledge, AI workflows, and integrated agent evaluation, built around a shared backend control plane.

- Designed a Django/DRF control plane for projects, permissions, document versions, processing jobs, artifacts, evaluation runs, scorecards, and audit metadata.
- Built versioned document-ingestion workflows covering source upload, OCR, preview generation, chunking, indexing workflows, committed artifacts, and S3-compatible artifact lineage.
- Defined service ownership through typed worker contracts, idempotent jobs, and concurrency-safe commits, keeping domain lifecycle decisions in the API while stateless Celery workers handle long-running processing.
- Integrated Agent Testing into the platform core with runtime trace capture, multi-stage evaluation, deterministic validation, and workflow-level quality gates.

IQP — Intelligent Quality Platform

An enterprise Quality Engineering platform combining product knowledge, source-code intelligence, and AI-assisted retrieval to support engineering traceability and test analysis.

- Designed backend service boundaries for workspace management, graph ingestion and retrieval, code indexing, AI context access, and quality-analysis workflows.
- Implemented hybrid retrieval combining semantic, lexical, and graph-based signals to provide scoped context for engineering and AI-assisted tools.
- Developed code-intelligence capabilities for indexing source repositories and linking product concepts with related implementation and test assets.
- Integrated asynchronous backend services, a React/TypeScript portal, and MCP-compatible AI clients into a workspace-oriented product workflow.

Core Banking & Payment Hub QA Automation

Client delivery work covering omnichannel banking and financial transaction integrations across API, database, regression, and automation testing.

- Built and demonstrated an omnichannel automation proof of concept across Web, iOS, and Android, contributing to approximately 20 person-months of follow-on delivery work.
- Validated end-to-end financial transactions through API, SQL-based database, regression, and stress testing, including more than 1,500 lines of reusable SQL validation queries.
- Reduced manual QA effort by approximately 70% through reusable automation and transaction-validation workflows.

APIT — Agent Programmatic Intelligence Testing

A graduation project that analyzes API documentation and generates structured API test cases using LLMs.

- Built a FastAPI-based workflow for API-document understanding, structured test-case generation, and automated output evaluation using LLM-as-a-Judge.
- Evaluated LoRA and QLoRA approaches and fine-tuned Qwen2.5-3B, achieving macro-F1 0.655, a 41.5% relative improvement over a Llama-3.2-3B baseline in the same evaluation setup.

Technical Skills

Backend Systems & Architecture: Python, Django/DRF, FastAPI, Pydantic, SQLAlchemy/SQLModel, REST APIs, domain modeling, service boundaries, asynchronous orchestration, idempotency, PostgreSQL, Redis, Celery

Applied AI & Graph Systems: RAG, hybrid retrieval, document intelligence, Knowledge Graphs, Code Graphs, graph traversal, agentic workflows, MCP, LangChain, LangGraph, LiteLLM, Docling, Qdrant, Langfuse

Product Interfaces & Delivery: React, TypeScript, Vite, Docker, Docker Compose, Nginx, CI/CD, observability, S3/MinIO, Git

Quality, Evaluation & ML: LLM evaluation, agent evaluation, LLM-as-a-Judge, quality gates, PyTorch, LoRA/QLoRA, model evaluation, API validation, SQL validation, regression testing, test automation

Education

FPT School of Business & Technology, Master of Software Engineering in AI — Ho Chi Minh City, Vietnam
May 2026 – present

FPT University, Bachelor of Artificial Intelligence — Ho Chi Minh City, Vietnam
2021 – 2025

Languages

Vietnamese: Native

English: Working proficiency